

COMPARATIVE STUDY OF STOCHASTIC MODELS FOR OPTION PRICING

In the Indian Stock Market (NIFTY 50)

Black-Scholes → Merton Jump-Diffusion → Heston → Bates

Project Report | Mid-Term Evaluation | Academic Year 2025-26

Under the supervision of Prof. Chaman Kumar

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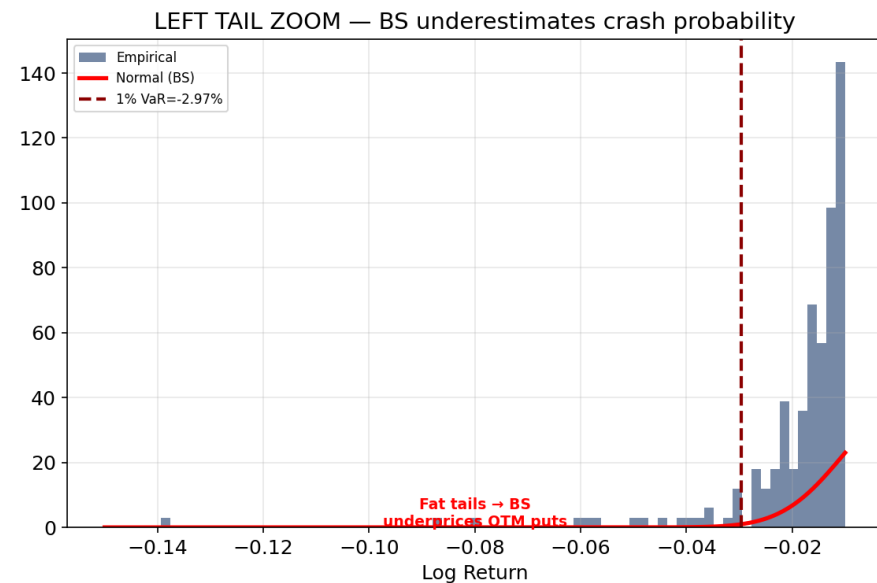
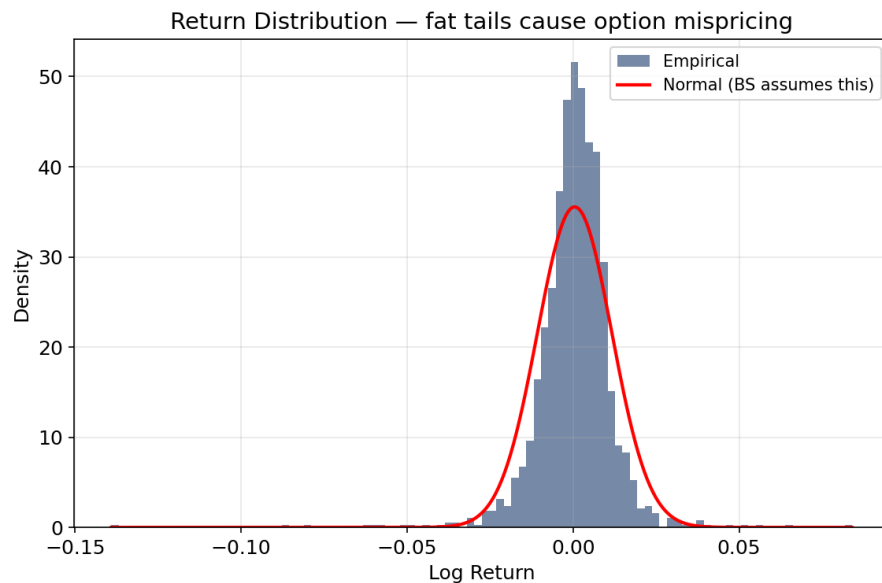
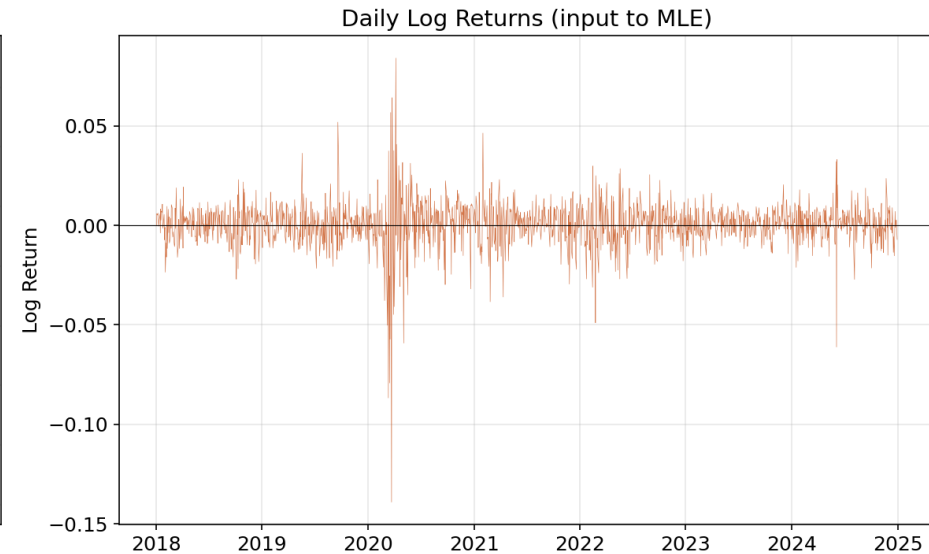
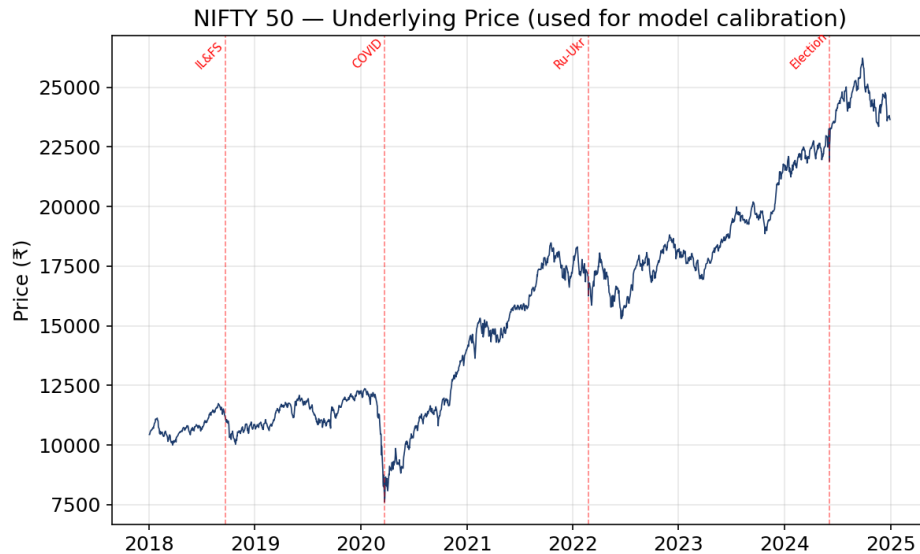
Outline

1. Motivation: why option pricing needs better models
2. Data & Model Calibration (MLE on NIFTY returns)
3. Black–Scholes option pricing:
 - GBM SDE $\rightarrow$ MLE $\rightarrow$ closed-form call/put $\rightarrow$ MC
4. Merton Jump–Diffusion option pricing:
 - SDE with jumps $\rightarrow$ MLE $\rightarrow$ MC pricing
5. Option price comparison:
 - Calls & puts across strikes
 - OTM puts: where BS fails catastrophically
 - Implied volatility smile
 - Option pricing at crisis events
6. Statistical model selection (LRT, AIC/BIC)
7. Future work: Heston & Bates models

1. Why Option Pricing Needs Better Models

- NIFTY 50 index options are among the most actively traded contracts globally
- Practitioners use Black-Scholes to price & hedge — but it systematically misprices:
 - Deep out-of-the-money (OTM) PUTS are underpriced → inadequate crash protection
 - The "volatility smile" in market data contradicts BS flat-IV assumption
- Root cause: BS assumes Gaussian log-returns (thin tails + constant volatility)
 - assigns near-zero probability to large moves like COVID -13% day, election -8% gap
- This project: compare BS vs Merton Jump-Diffusion option prices on NIFTY 50
 - calibrate both models to historical data, then price European calls & puts
 - quantify how much BS underprices crash protection
- References: Black & Scholes (1973), Merton (1976)

2. Calibration Data: NIFTY 50 (2018–2024)



NIFTY daily returns used to estimate model parameters via MLE. Left-tail zoom shows fat tails → BS underprices OTM puts.

Source: Yahoo Finance (^NSEI). Risk-free rate: RBI repo $\approx$ 6.5%

3. Black–Scholes: GBM SDE & Log-Price

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t)$$

$$d(\ln S) = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW(t)$$

$$S(T) = S_0 \cdot \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T} Z\right], \quad Z \sim \mathcal{N}(0, 1)$$

$$r_t = \ln \frac{S_t}{S_{t-1}} \sim \mathcal{N}\left(\left(\mu - \frac{\sigma^2}{2}\right)\Delta t, \sigma^2 \Delta t\right)$$

GBM SDE $\rightarrow$ Itô's lemma gives the log-price SDE.
Integrating gives the log-normal terminal distribution $S(T)$.
Daily log-returns are i.i.d. Gaussian — this is the model we fit via MLE.
The estimated $\hat{\sigma}$ is then plugged into the BS option pricing formula.

3. Black–Scholes PDE: Eliminating $dW(t)$ via Delta Hedging (Proof)

$$\Pi = V(S, t) - \Delta S, \quad d\Pi = dV - \Delta dS$$

$$dV = \left(V_t + \mu S V_S + \frac{1}{2} \sigma^2 S^2 V_{SS} \right) dt + \sigma S V_S dW$$

$$\Delta = V_S \Rightarrow \sigma S (V_S - \Delta) dW = 0, \quad \mu \text{ cancels}$$

$$d\Pi = r_f \Pi dt \Rightarrow V_t + \frac{1}{2} \sigma^2 S^2 V_{SS} + r_f S V_S - r_f V = 0$$

Portfolio Π = long 1 option, short Δ shares.

Apply Itô to $V(S, t)$; substitute $dS = \mu S dt + \sigma S dW$.

Choose $\Delta = V_S \Rightarrow$ the random term cancels: dW disappears.

Crucial: μ cancels too $\Rightarrow$ option price does NOT depend on expected stock return.

No-arbitrage: riskless Π must earn $r_f \Rightarrow$ Black–Scholes PDE.

This is the key Nobel-Prize insight: dynamic replication $\rightarrow$ risk-neutral pricing.

3. Black–Scholes: Maximum Likelihood Estimation

$$\ell(\mu, \sigma) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2 \Delta t) - \sum_{i=1}^n \frac{(r_i - \tilde{\mu}\Delta t)^2}{2\sigma^2 \Delta t}$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n\Delta t} \sum_{i=1}^n (r_i - \bar{r})^2, \quad \hat{\mu}_{MLE} = \frac{\bar{r}}{\Delta t} + \frac{\hat{\sigma}^2}{2}$$

$$dS(t) = r_f S(t) dt + \sigma S(t) dW^{\mathbb{Q}}(t)$$

MLE on n daily NIFTY returns. Closed-form for $\hat{\sigma}^2$ and $\hat{\mu}$ exists for GBM.

We use numerical optimisation (Nelder-Mead) for consistency with MJD estimation.

The key output for option pricing is $\hat{\sigma} \approx 17.8\%$ (annualised).

For pricing: switch to risk-neutral measure $\mathbb{Q}$. Drift becomes r_f , but σ stays the same.

3.1 MLE for Discretely Sampled SDEs (Why it is hard)

$$\ell_n(\theta) = \sum_{i=1}^n \ln \{p(\Delta, x_i \mid x_{i-1}; \theta)\}$$

Discrete data: $X_{\{t_i\}}$ observed at fixed Δ (e.g., daily).
Exact MLE needs the transition density $p(\Delta, x_i \mid x_{i-1}; \theta)$.
For most nonlinear SDEs this density is NOT available in closed form.

Aït-Sahalia's key statement (Econometrica): for discretely sampled diffusions,
“the transition distribution and hence the likelihood ... is not explicitly computable” in most cases,
so he constructs a convergent closed-form approximation using Hermite polynomials. [AS02]

$$p(\Delta, x \mid x_0; \theta) \approx p^{(J)}(\Delta, x \mid x_0; \theta) \quad (\text{closed-form Hermite expansion, } J=2, 3)$$

Iacus (Springer book) formalizes the “fixed- Δ ” scheme and emphasizes that with discrete-time observations the likelihood is “almost always unavailable in explicit form,” motivating pseudo/approximated likelihoods. [Iac08]

3. Black–Scholes: Closed-Form Option Prices

$$C = S_0 \Phi(d_1) - K e^{-r_f T} \Phi(d_2)$$

$$d_1 = \frac{\ln(S_0/K) + (r_f + \sigma^2/2) T}{\sigma \sqrt{T}}, \quad d_2 = d_1 - \sigma \sqrt{T}$$

$$\hat{C}_{MC} = e^{-r_f T} \cdot \frac{1}{N} \sum_{i=1}^N \max(S_T^{(i)} - K, 0)$$

The celebrated BS formula gives exact European call (and put via put-call parity). d_1 and d_2 encode moneyness, time to expiry, and volatility.

Monte Carlo estimator validates the closed-form: simulate N paths under Q , average discounted payoffs.

SE shrinks as $1/\sqrt{N}$ (CLT). [Glasserman, 2003]

Key limitation: σ is CONSTANT across all strikes and maturities $\rightarrow$ flat implied volatility surface.
 $\rightarrow$ In reality, NIFTY options trade with a pronounced volatility smile/skew. BS cannot reproduce this.

4. Merton Jump-Diffusion: SDE & Log-Return

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t) + S(t^-)(e^J - 1) dN(t)$$

$$N(t) \sim \text{Poisson}(\lambda), \quad J \sim \mathcal{N}(\mu_J, \sigma_J^2)$$

$$d(\ln S) = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW(t) + J dN(t)$$

$$r \mid N = k \sim \mathcal{N}\left(\left(\mu - \frac{\sigma^2}{2}\right) \Delta t + k\mu_J, \sigma^2 \Delta t + k\sigma_J^2\right)$$

Superimposes compound Poisson jumps on GBM — captures sudden dislocations.

Itô for jump-diffusions: extra $J dN(t)$ term in log-price. Conditional on k jumps: Gaussian with shifted mean & inflated variance.

This directly explains why MJD produces different option prices from BS.

[Me76] Merton (1976), [CT04] Cont & Tankov (2004).

4. Merton JD Option Pricing: Risk-Neutral Compensator & Series (Proof)

$$\kappa = \mathbb{E}[e^J - 1] = e^{\mu_J + \sigma_J^2/2} - 1$$

$$dS(t) = (r_f - \lambda\kappa) S(t^-)dt + \sigma S(t^-) dW^{\mathbb{Q}}(t) + S(t^-)(e^J - 1) dN(t)$$

$$\sigma_k^2 = \sigma^2 + \frac{k\sigma_J^2}{T}, \quad r_k = r_f - \lambda\kappa + \frac{k\mu_J}{T} + \frac{k\sigma_J^2}{2T}$$

$$V_{MJD} = \sum_{k=0}^{\infty} \frac{e^{-\lambda T} (\lambda T)^k}{k!} \text{BS}(S_0, K, T, r_k, \sigma_k)$$

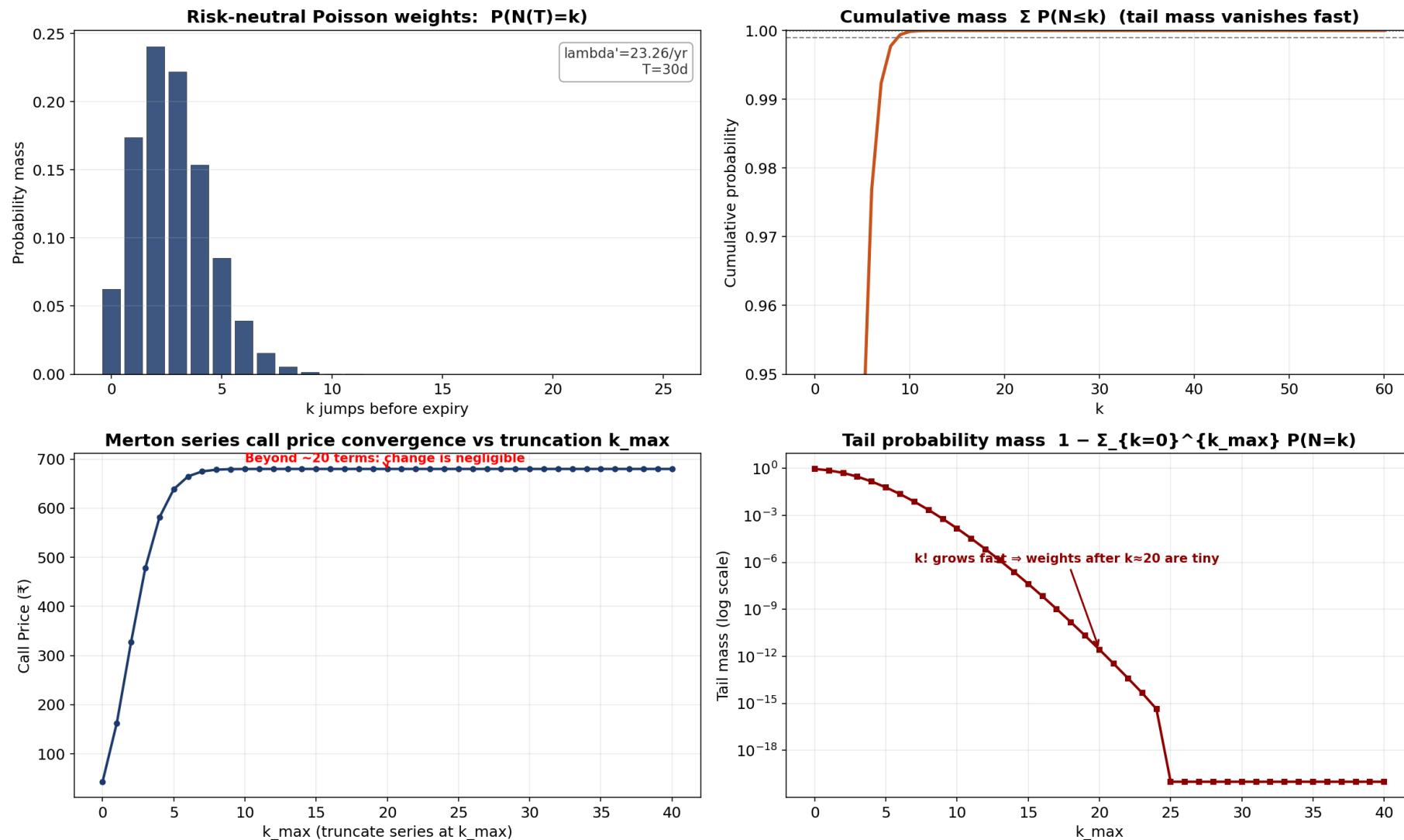
Key arbitrage constraint: under $\mathbb{Q}$, total expected growth (diffusion + expected jumps) must equal r_f .

Hence drift becomes $(r_f - \lambda\kappa)$. Conditional on k jumps, the terminal log-price is Gaussian $\Rightarrow$ BS formula with modified (r_k, σ_k) .

Final price is a Poisson-weighted average of BS prices across k . [Me76]

4.2 Why truncating the infinite series is safe (Visualization)

Why truncating the Merton infinite series is safe (Law of Total Expectation + Poisson tails)



Poisson weights decay super-fast due to $k!$ in the denominator $\Rightarrow$ tail mass after $k \approx 20$ is negligible; price stabilizes.

This justifies truncating the Merton series in numerical implementations.

4.1 MLE in Practice: Exact vs Pseudo/Approximated Likelihood (Iacus)

Iacus (2008) classifies estimation under discrete observations into:

- Exact likelihood inference (when transition density is known, e.g., GBM/BS)
- Pseudo-likelihood methods (e.g., Euler Gaussian approximation)
- Approximated likelihood methods (Kessler, simulated likelihood, Hermite expansion)

$$X_{t+\Delta} \approx X_t + \mu(X_t; \theta)\Delta + \sigma(X_t; \theta)\sqrt{\Delta} Z, \quad Z \sim N(0, 1)$$

Euler pseudo-likelihood uses the local Gaussian approximation over Δ :
this is the baseline idea behind many feasible estimators when $p(\Delta, \cdot | \cdot; \theta)$ is unknown.

In our project:

- BS/GBM has an explicit Gaussian transition for log-returns $\Rightarrow$ exact MLE is easy.
- MJD has a Poisson–Gaussian mixture transition $\Rightarrow$ we maximize a (truncated) mixture log-likelihood numerically.

Reference: Iacus (2008), Chapter 3 (Parametric Estimation). [Iac08]

4. MJD: Likelihood & Excess Kurtosis

$$f(r) = \sum_{k=0}^{\infty} \frac{e^{-\lambda \Delta t} (\lambda \Delta t)^k}{k!} \cdot \varphi(r; \tilde{\mu} \Delta t + k \mu_J, \sigma^2 \Delta t + k \sigma_J^2)$$

$$\ell(\theta) = \sum_{t=1}^n \ln \left[\sum_{k=0}^K \frac{e^{-\lambda \Delta t} (\lambda \Delta t)^k}{k!} \cdot \varphi(r_t; m_k, v_k) \right]$$

$$\text{Var}(r) = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2)$$

$$\text{Kurt}_{MJD} = \frac{3\lambda (\mu_J^2 + \sigma_J^2)^2}{(\sigma^2 + \lambda(\mu_J^2 + \sigma_J^2))^2}$$

Log-likelihood = sum of log Gaussian-mixture densities.

Maximised numerically (L-BFGS-B) over $\theta = (\mu, \sigma, \lambda, \mu_J, \sigma_J)$.

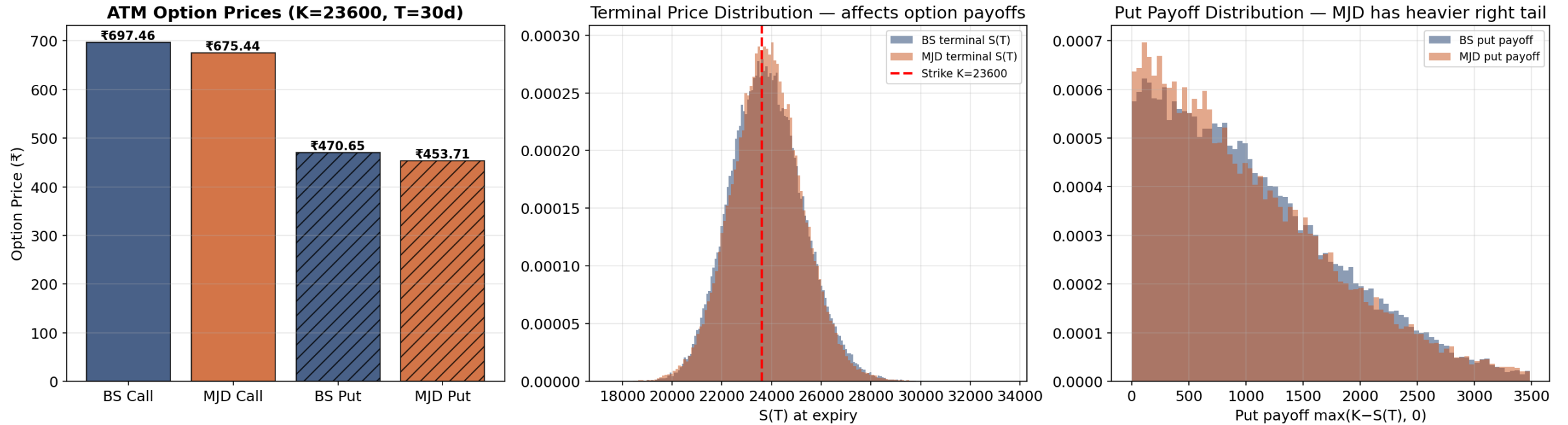
Variance decomposition: jump component $\lambda(\mu_J^2 + \sigma_J^2)$ adds kurtosis.

Excess kurtosis $> 0 \rightarrow$ fatter tails $\rightarrow$ higher OTM option prices.

This is the precise mechanism by which MJD prices puts higher than BS.

5. ATM Option Prices: BS vs MJD

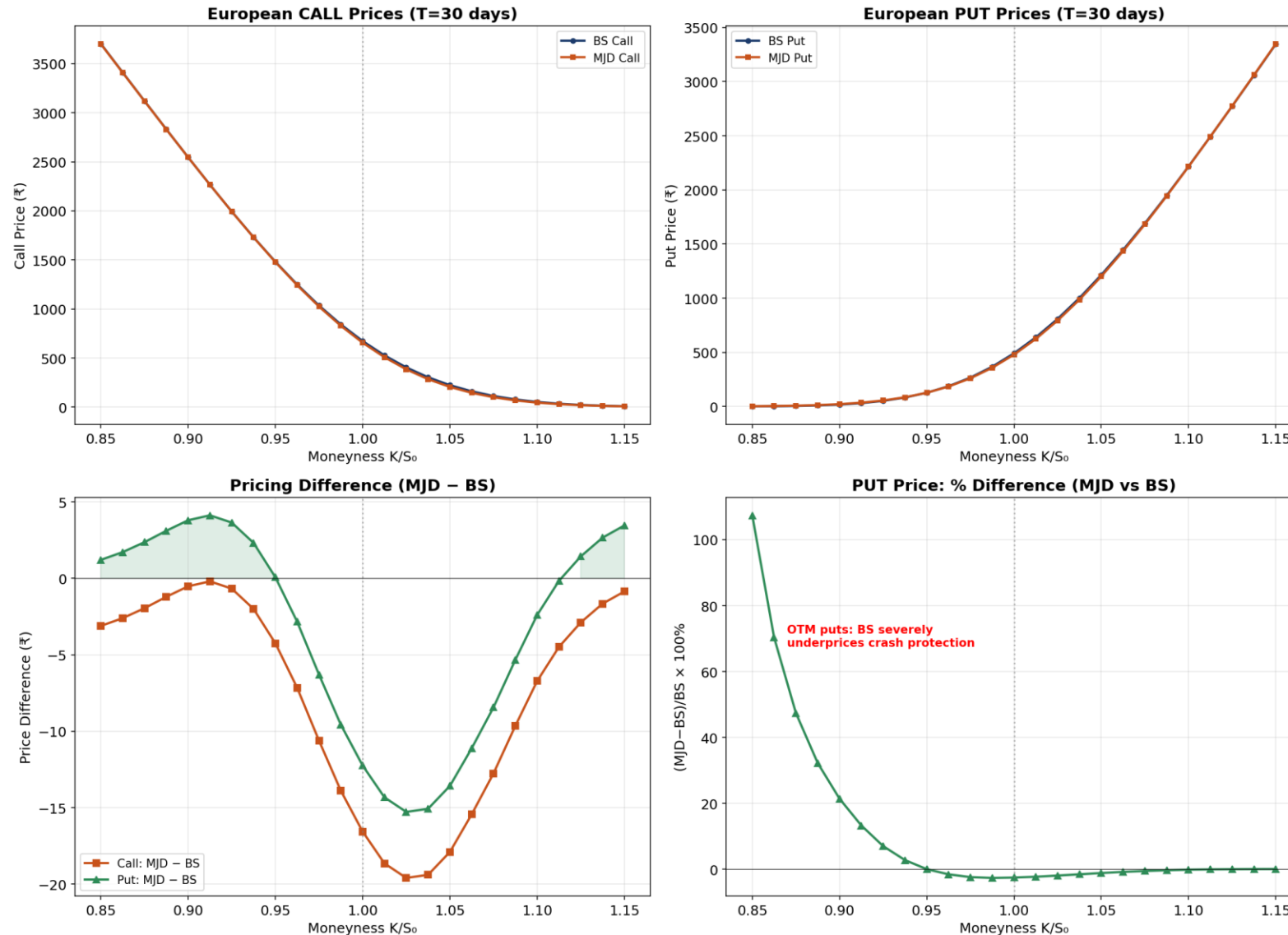
Option Pricing Comparison: Black-Scholes vs Merton Jump-Diffusion



Left: ATM call & put prices. Centre: terminal $S(T)$ distributions — MJD has heavier tails.
Right: put payoff distribution — MJD assigns more weight to large payoffs (crash scenarios).

5. Option Prices Across Strikes

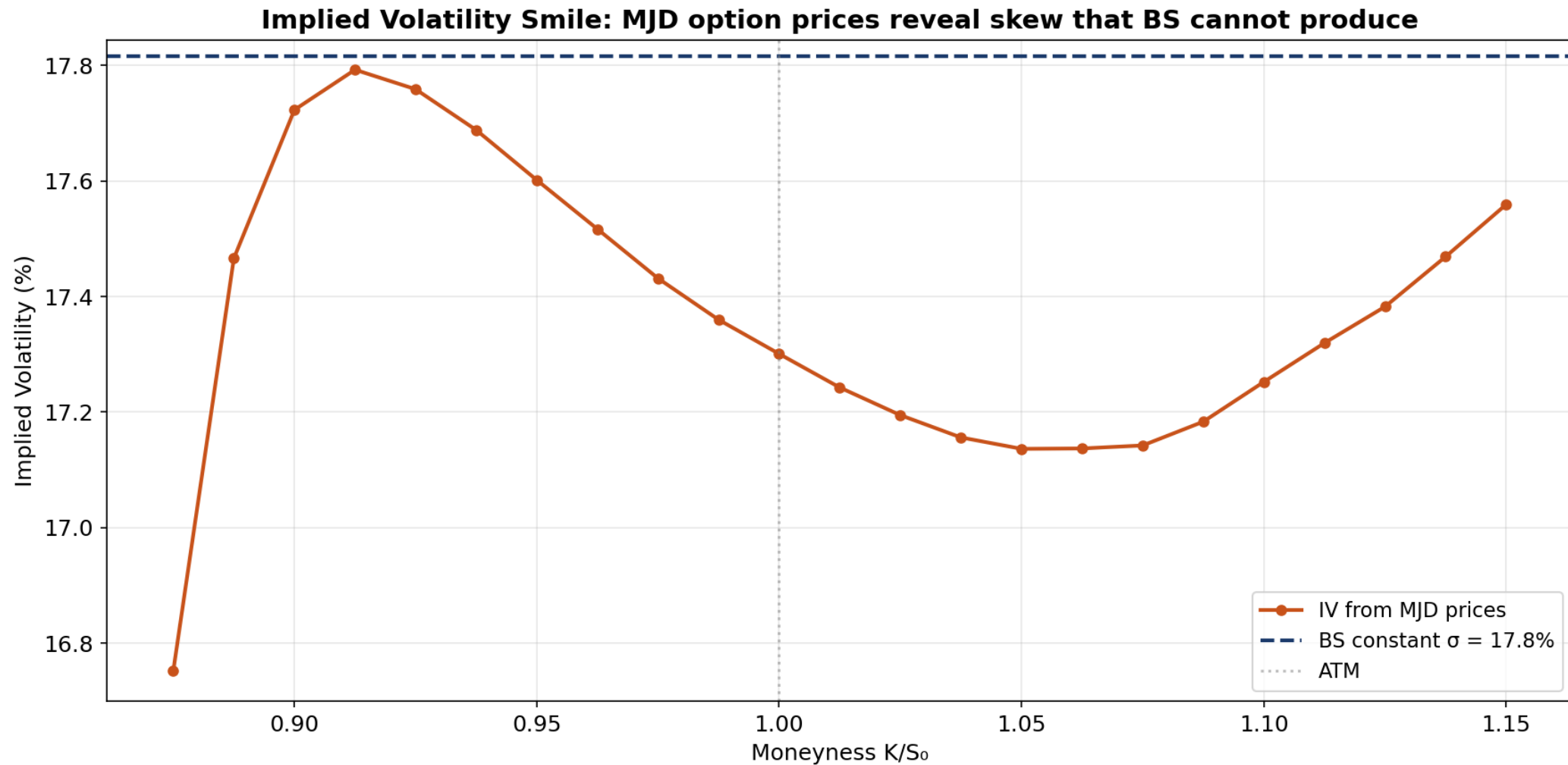
Option Prices Across Strikes: BS vs MJD (T=30 days)



MJD puts are consistently more expensive than BS puts, especially for OTM puts ($K/S_0 < 1$).

Bottom-right: percentage difference shows BS can underprice OTM puts by 20-100%+.

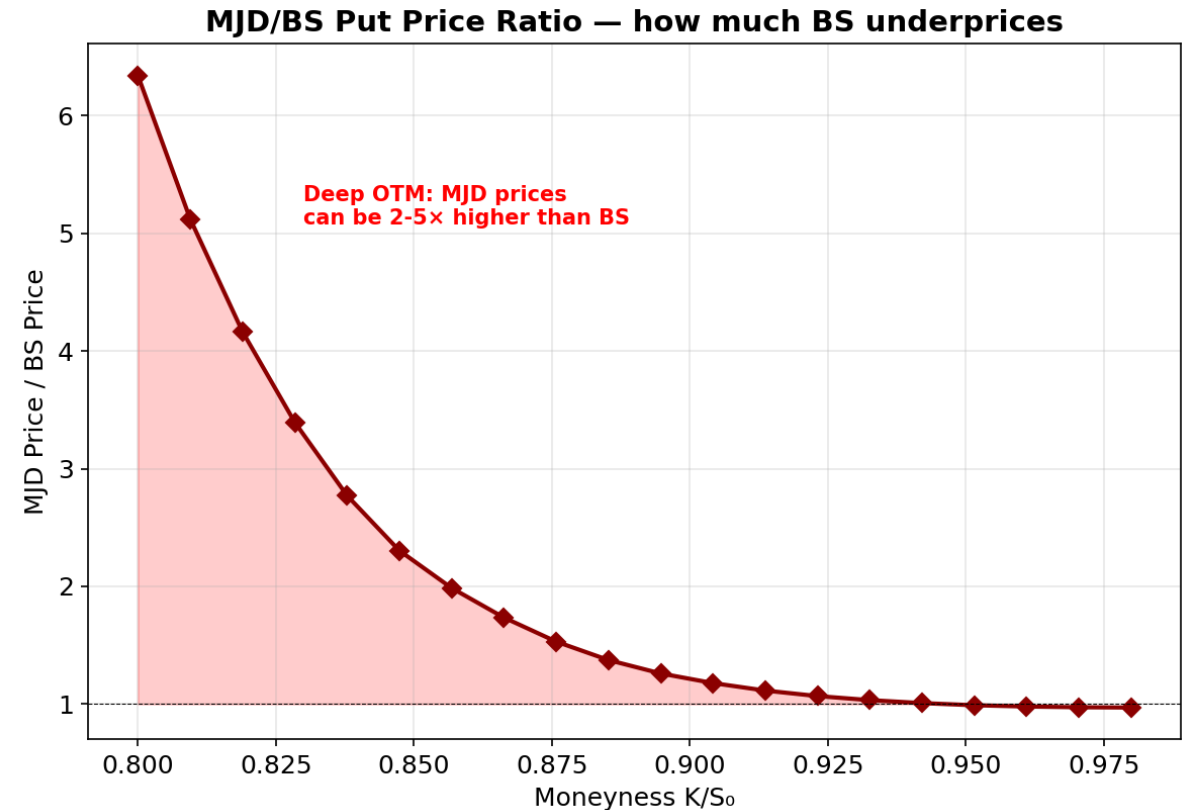
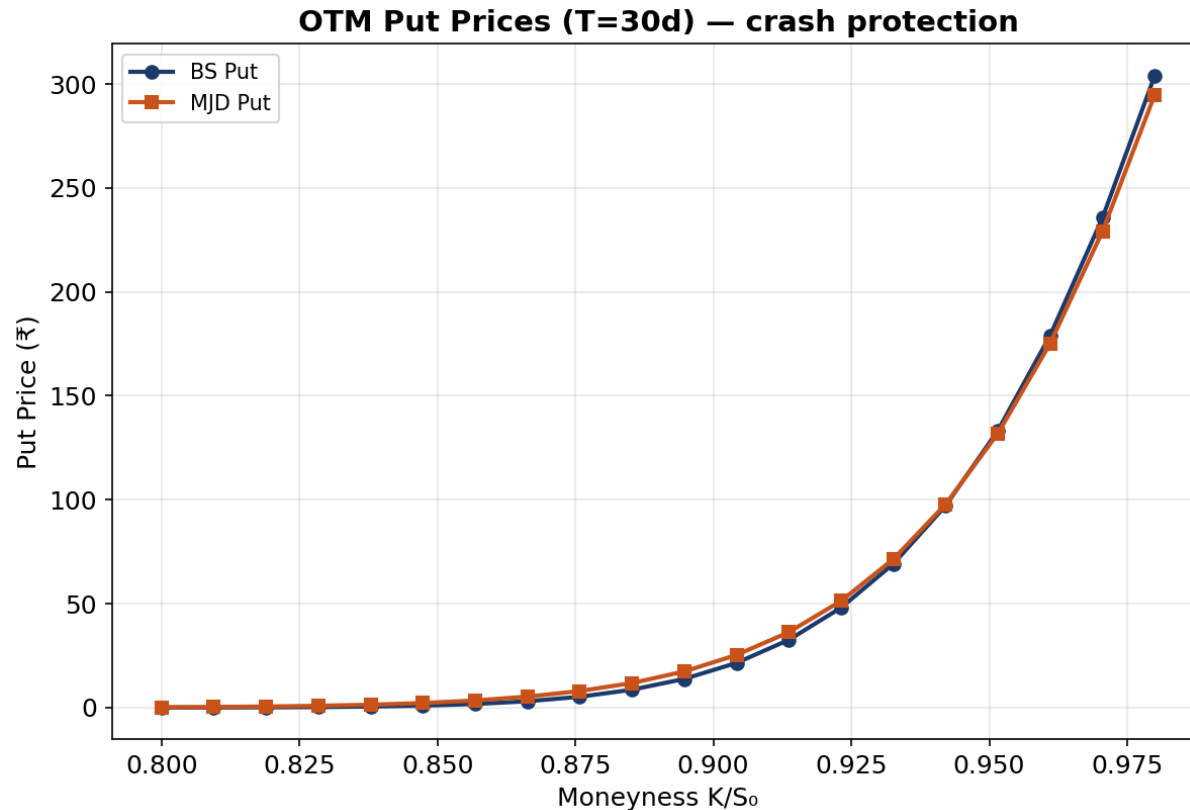
5. Implied Volatility Smile



BS-implied volatility backed out from MJD option prices. MJD produces the smile that market data shows. BS can only produce a flat line — the "smile inconsistency" [Rubinstein, 1994].

5. OTM Put Pricing — Where BS Fails Most

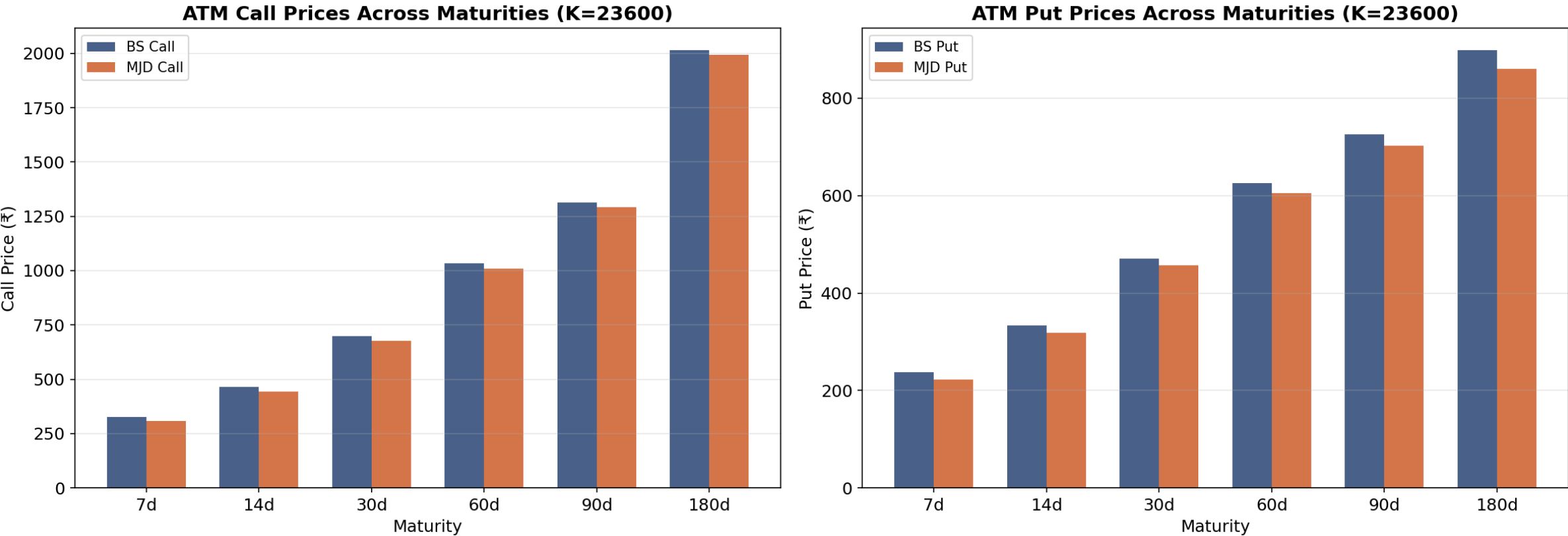
OTM Put Option Pricing: BS systematically underprices crash protection



Deep OTM puts: MJD price can be 2–5× the BS price. This is crash protection that BS undervalues. For a NIFTY options trader, using BS means systematically selling insurance too cheaply.

5. Option Prices Across Maturities

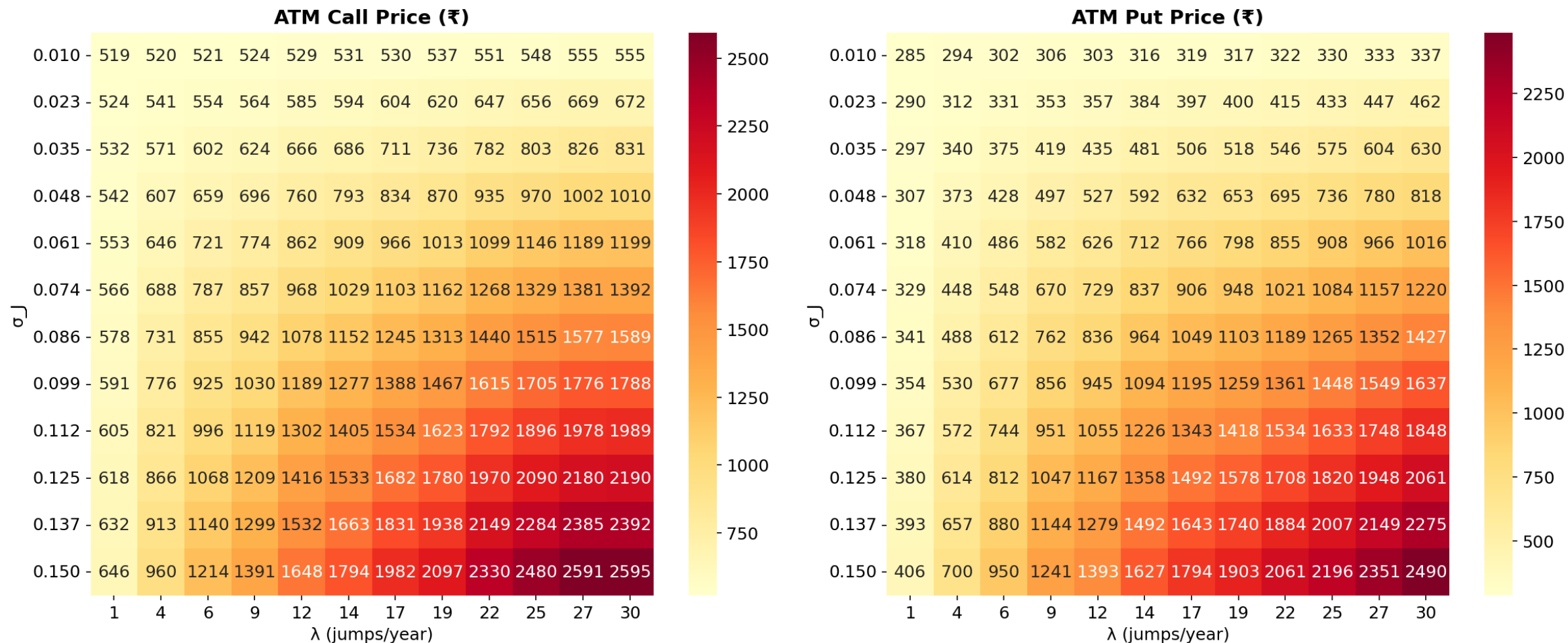
Option Price Term Structure: MJD puts are consistently more expensive (crash risk premium)



MJD puts are consistently more expensive than BS across all maturities — the jump risk premium persists. Difference is largest for shorter maturities where individual jumps have more impact.

5. Option Price Sensitivity to Jump Parameters

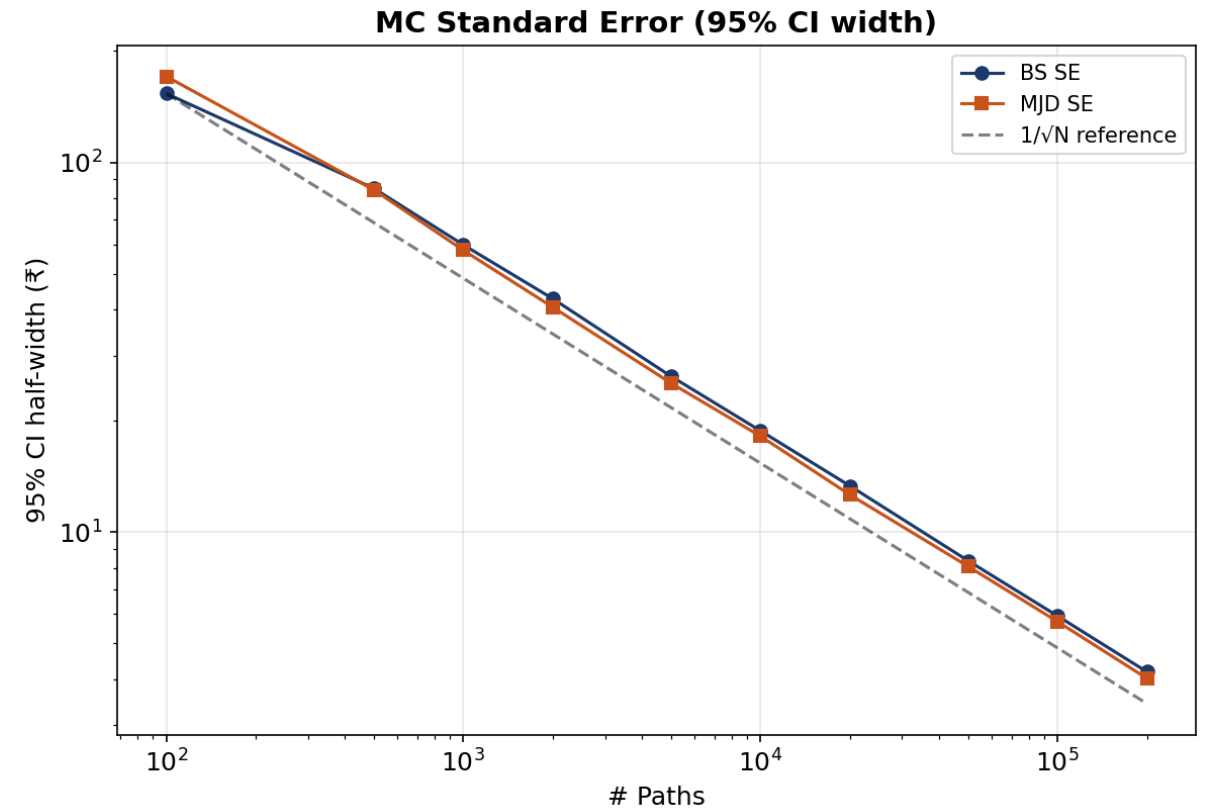
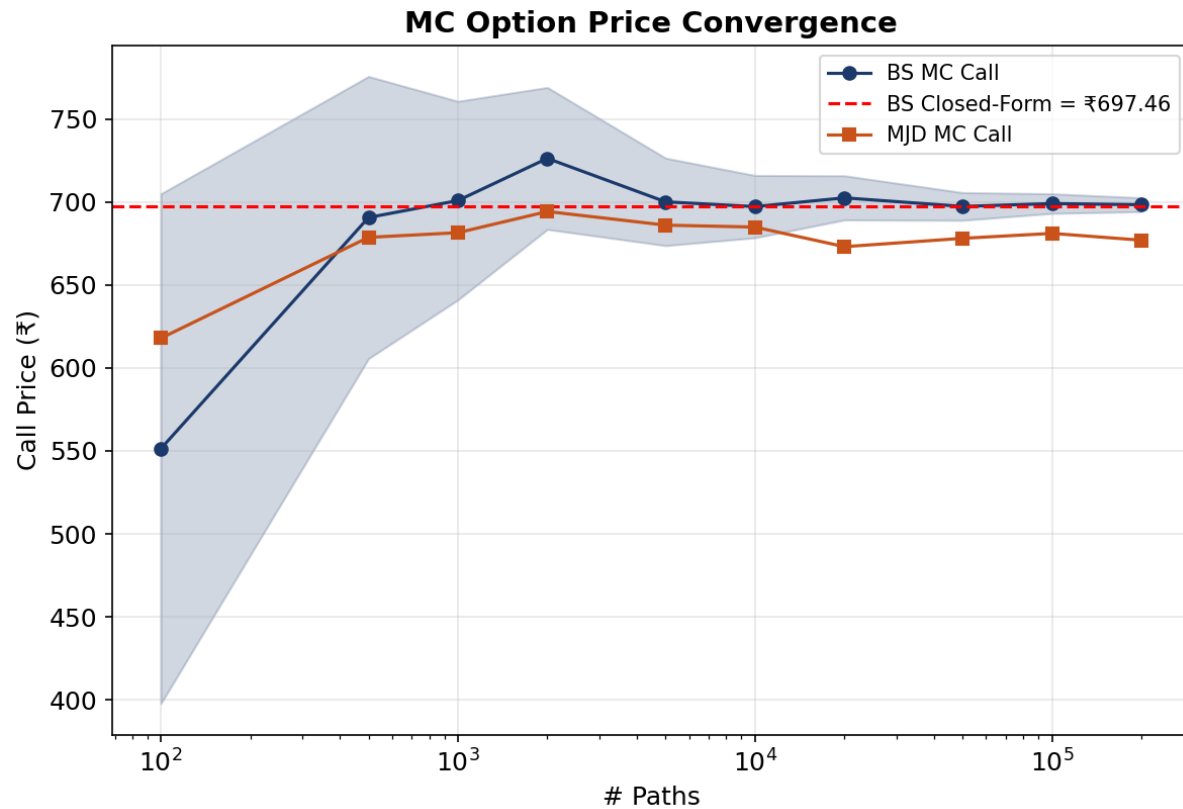
MJD Option Price Sensitivity to Jump Parameters



Non-linear interaction between λ and σ_J on option prices. Higher jump intensity and vol $\rightarrow$ higher option prices.

5. Monte Carlo Convergence (Option Prices)

Monte Carlo Convergence for Option Pricing

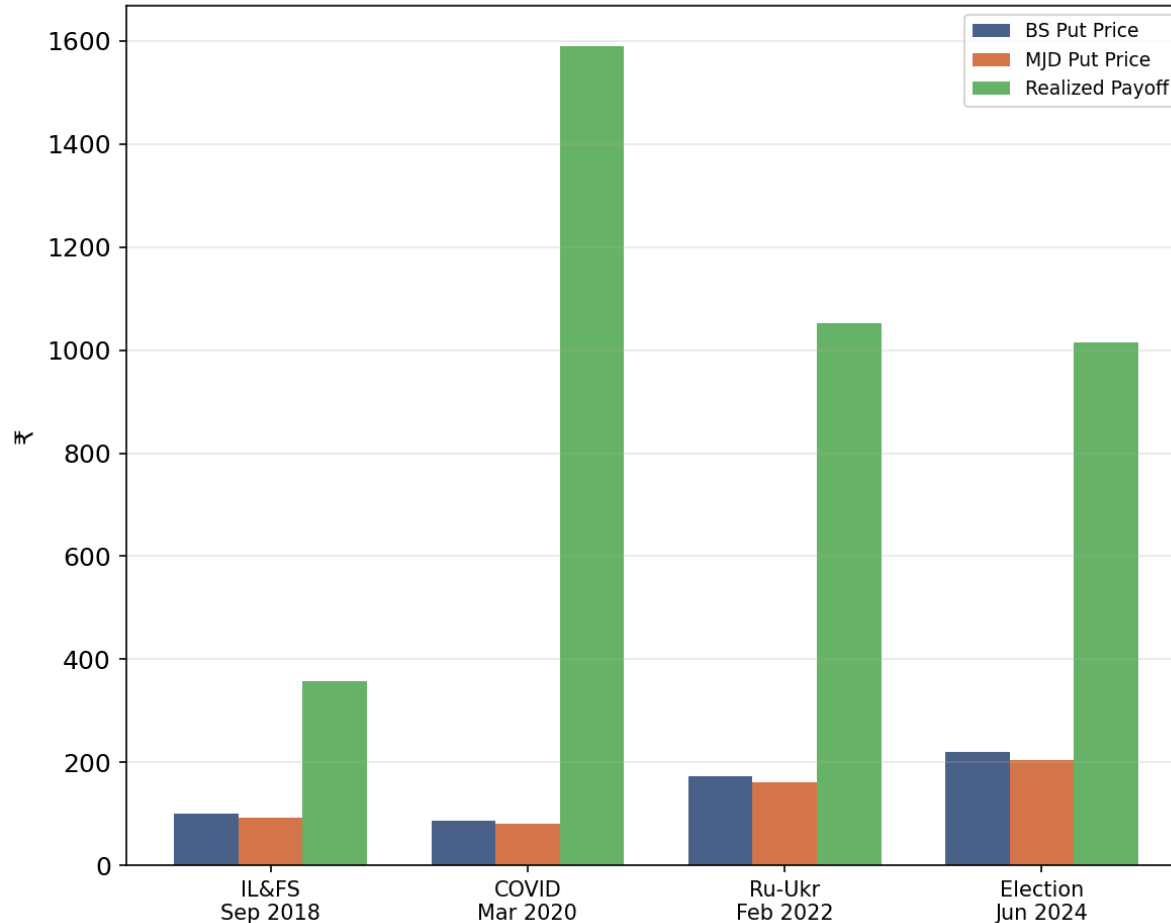


BS MC converges to closed-form; MJD converges to its own value. $SE \propto 1/\sqrt{N}$ confirmed.
Validates both implementations and the theoretical pricing framework [Glasserman, 2003].

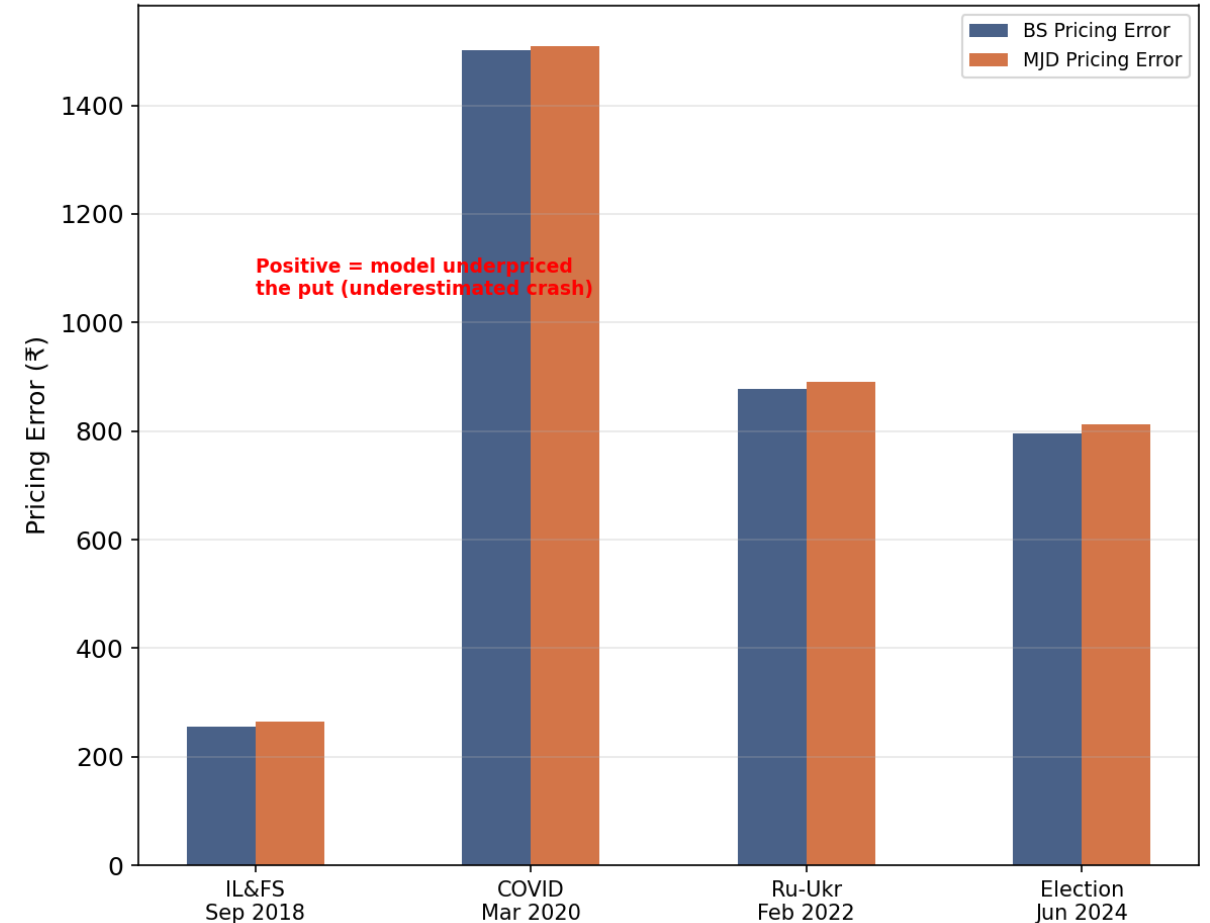
5. Option Pricing at Indian Market Crises

Option Pricing at Indian Market Crises: BS consistently underprices protective puts

ATM Put: Model Price vs Realized Payoff at Events

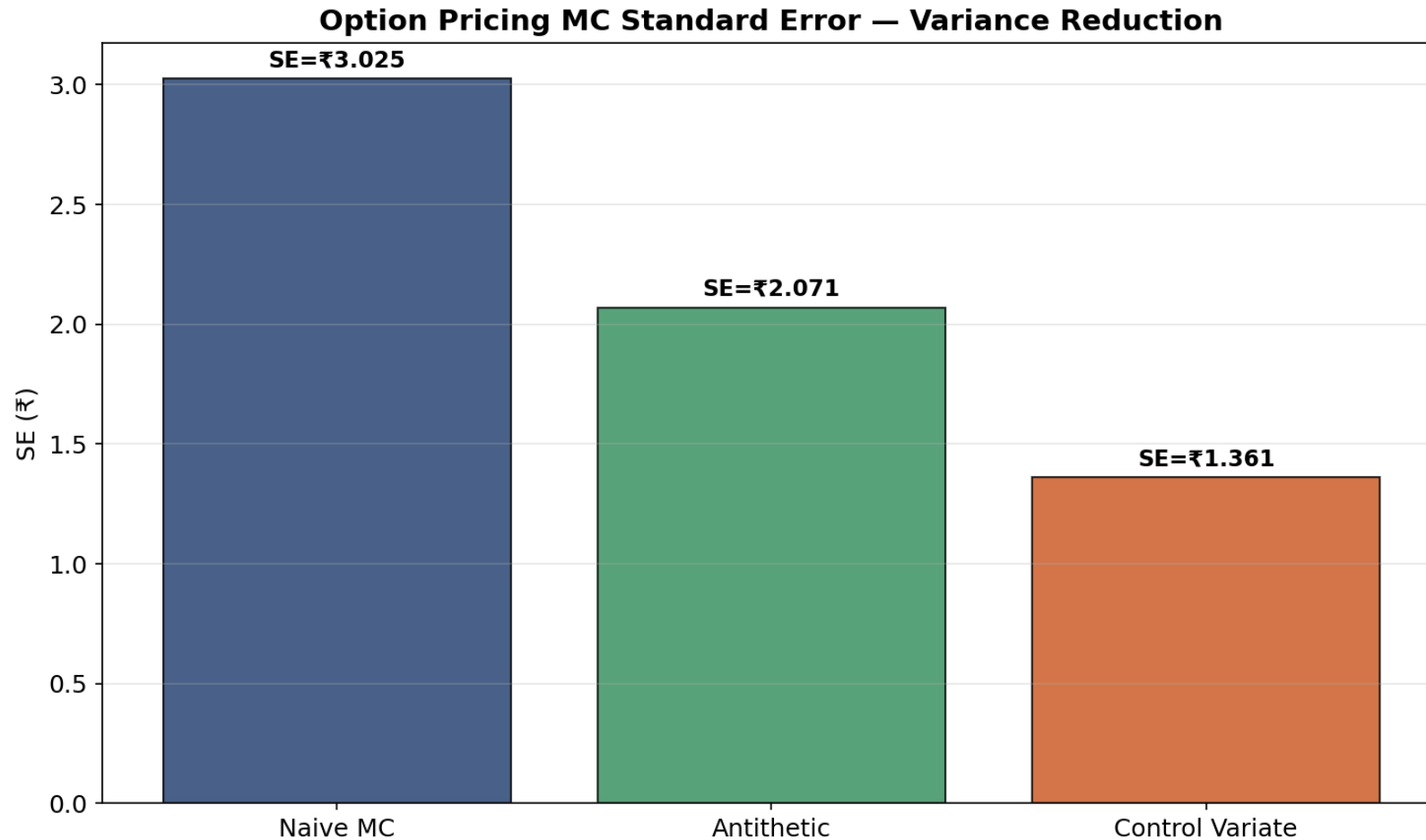


Option Pricing Error = Realized – Model Price



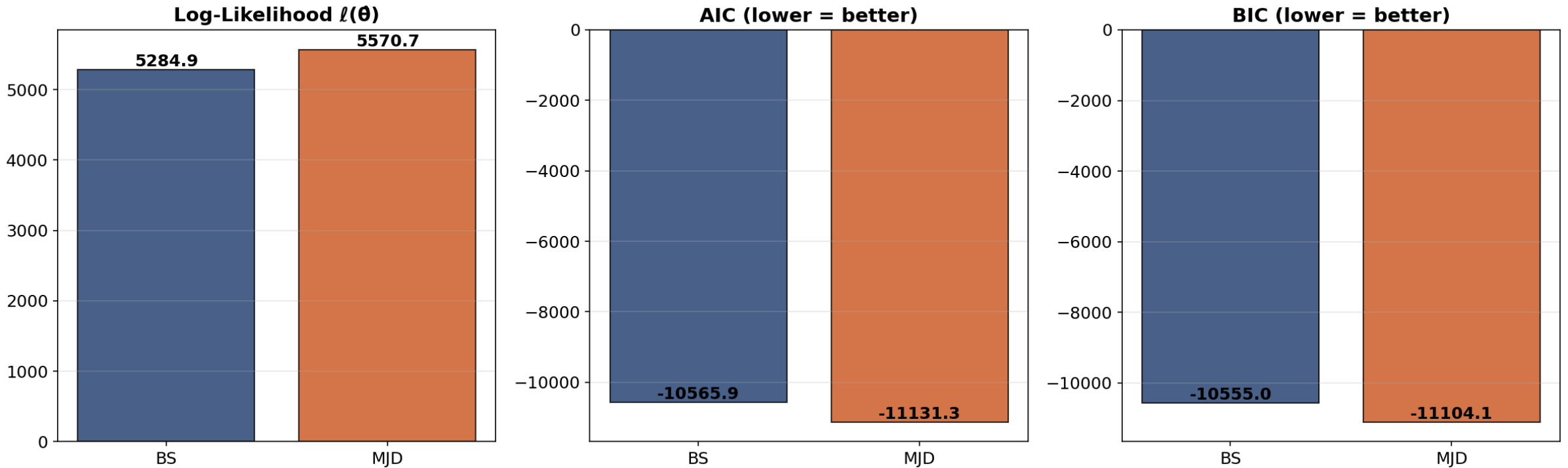
At each crisis: BS put price vs MJD put price vs realised payoff. BS consistently underprices — positive error = underpricing.

5. Variance Reduction for Option Pricing MC



Antithetic variates and control variates reduce MC standard error for option prices. Control variate (using BS closed-form as control for MJD MC) gives ~50% SE reduction. [Glasserman, 2003].

6. Statistical Model Selection

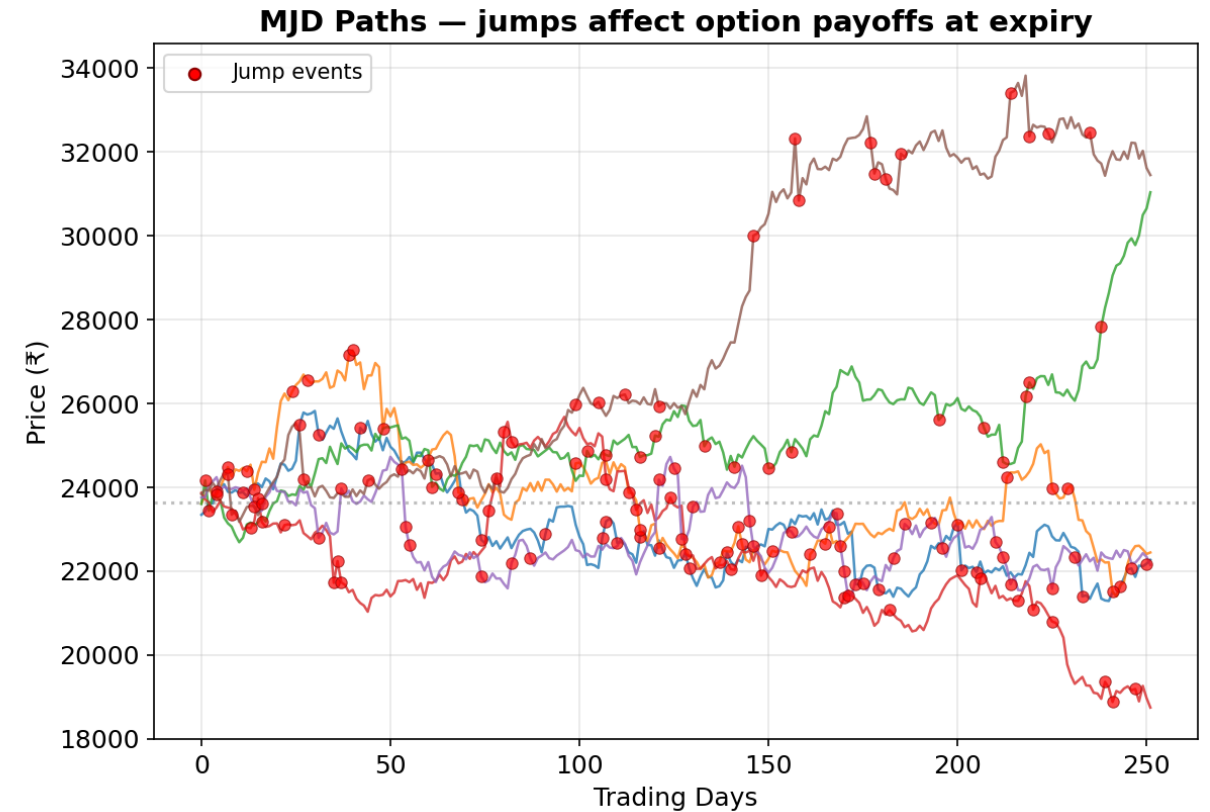
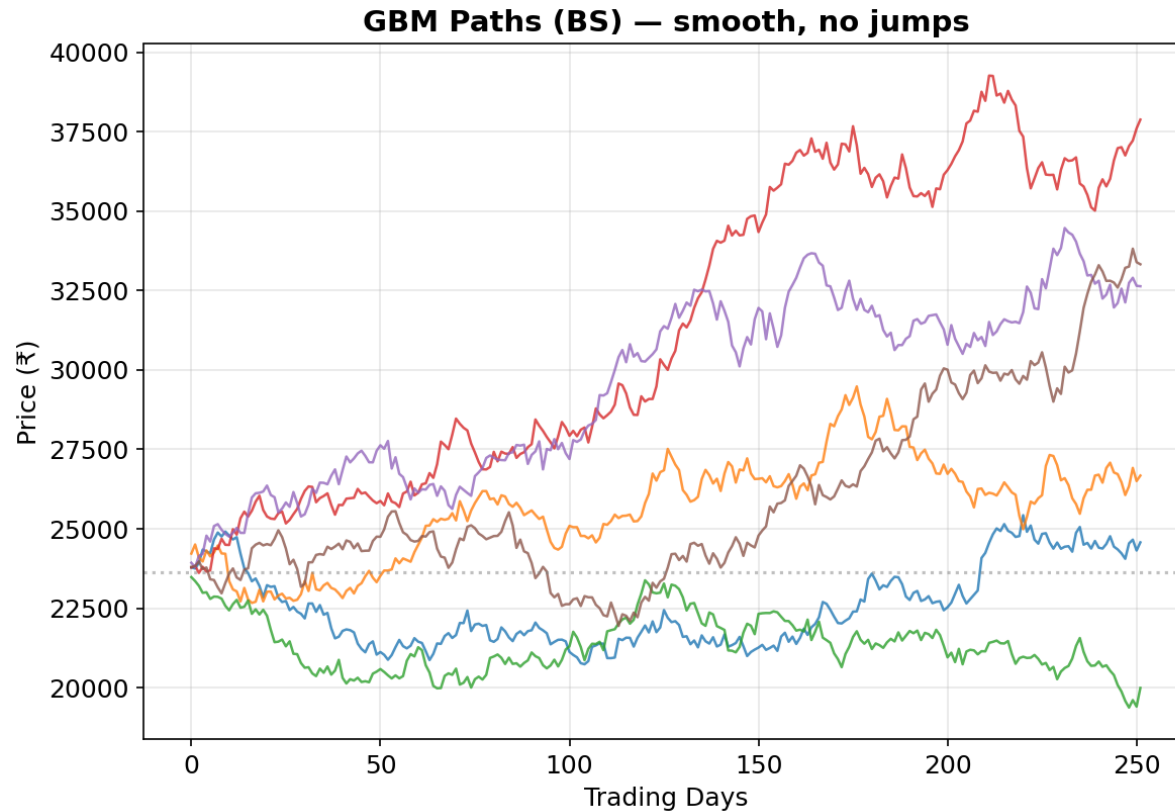


$$\Lambda = 2[\ell(\hat{\theta}_{MJD}) - \ell(\hat{\theta}_{BS})] \sim \chi^2(3) \text{ under } H_0 : \lambda = 0 \quad \text{AIC} = 2k - 2\ell(\hat{\theta}), \quad \text{BIC} = k\ln(n) - 2\ell(\hat{\theta})$$

LRT p-value $\approx 0 \rightarrow$ jumps are overwhelmingly significant. AIC & BIC both prefer MJD despite penalty for 3 extra parameters.

Underlying Paths: How Jumps Affect $S(T)$

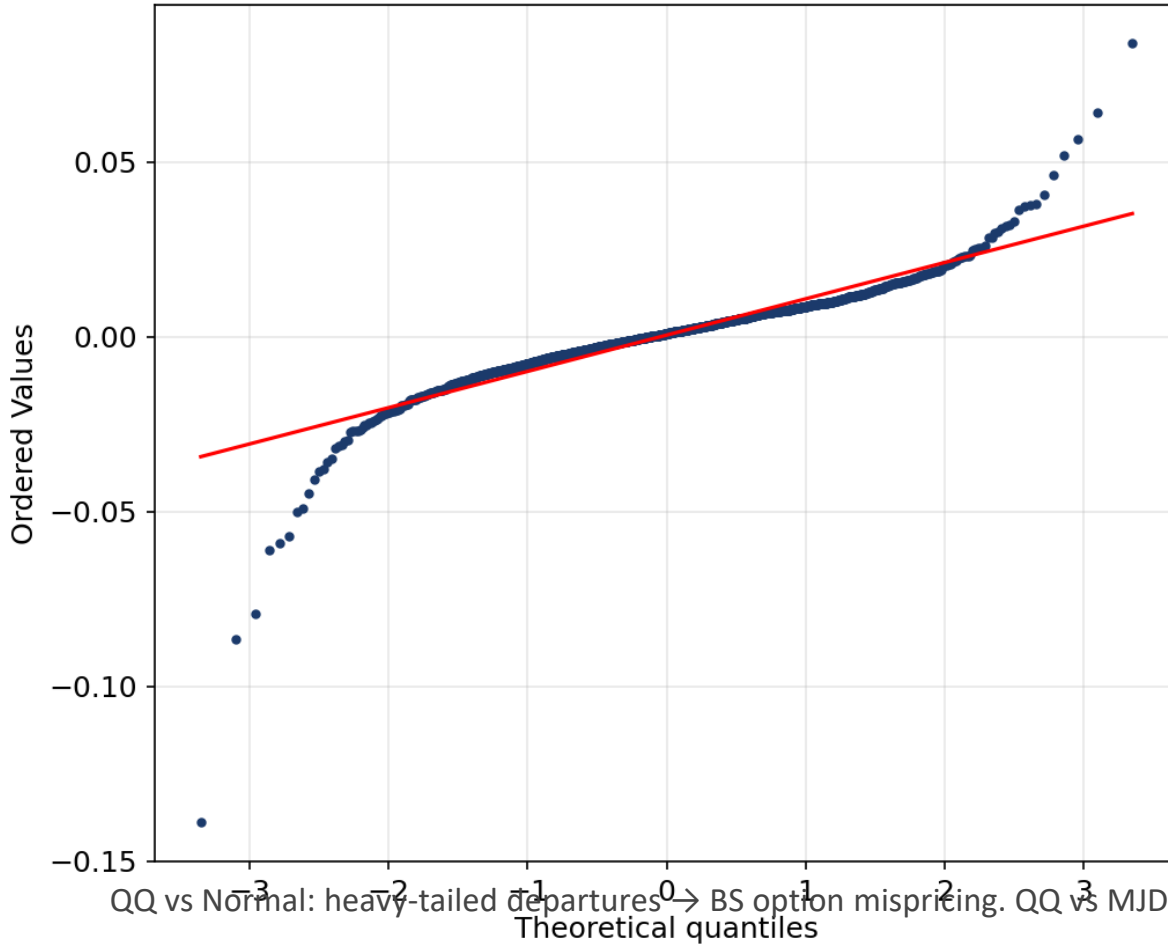
Underlying Price Paths: jumps create fat tails in $S(T)$ → affect option prices



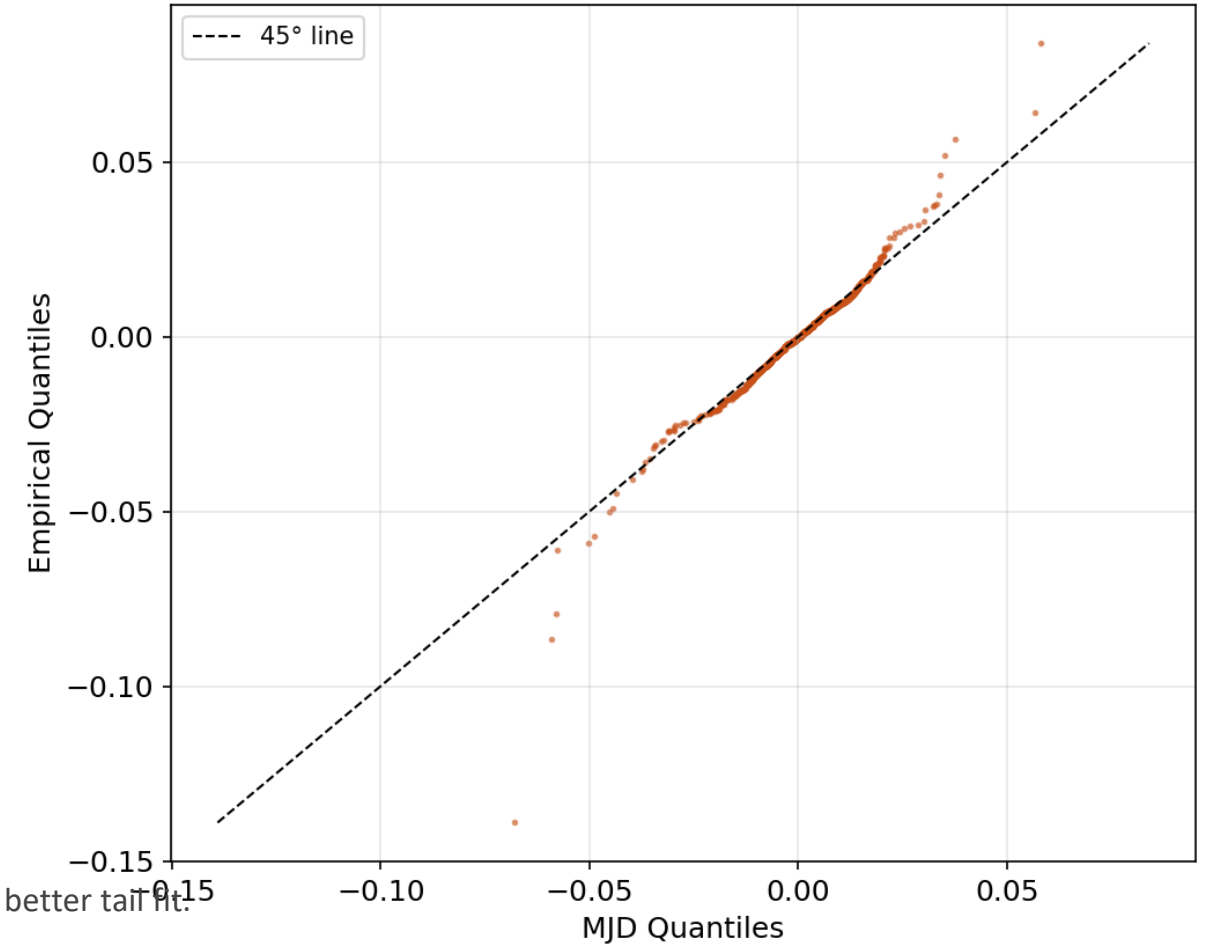
GBM paths are smooth; MJD paths have jumps (red dots). Jump events change the terminal distribution $S(T)$, which is what determines option payoffs $\max(S(T) - K, 0)$ or $\max(K - S(T), 0)$.

Return Diagnostics (Motivation for MJD)

QQ: Empirical vs Normal — tail departure → option mispricing

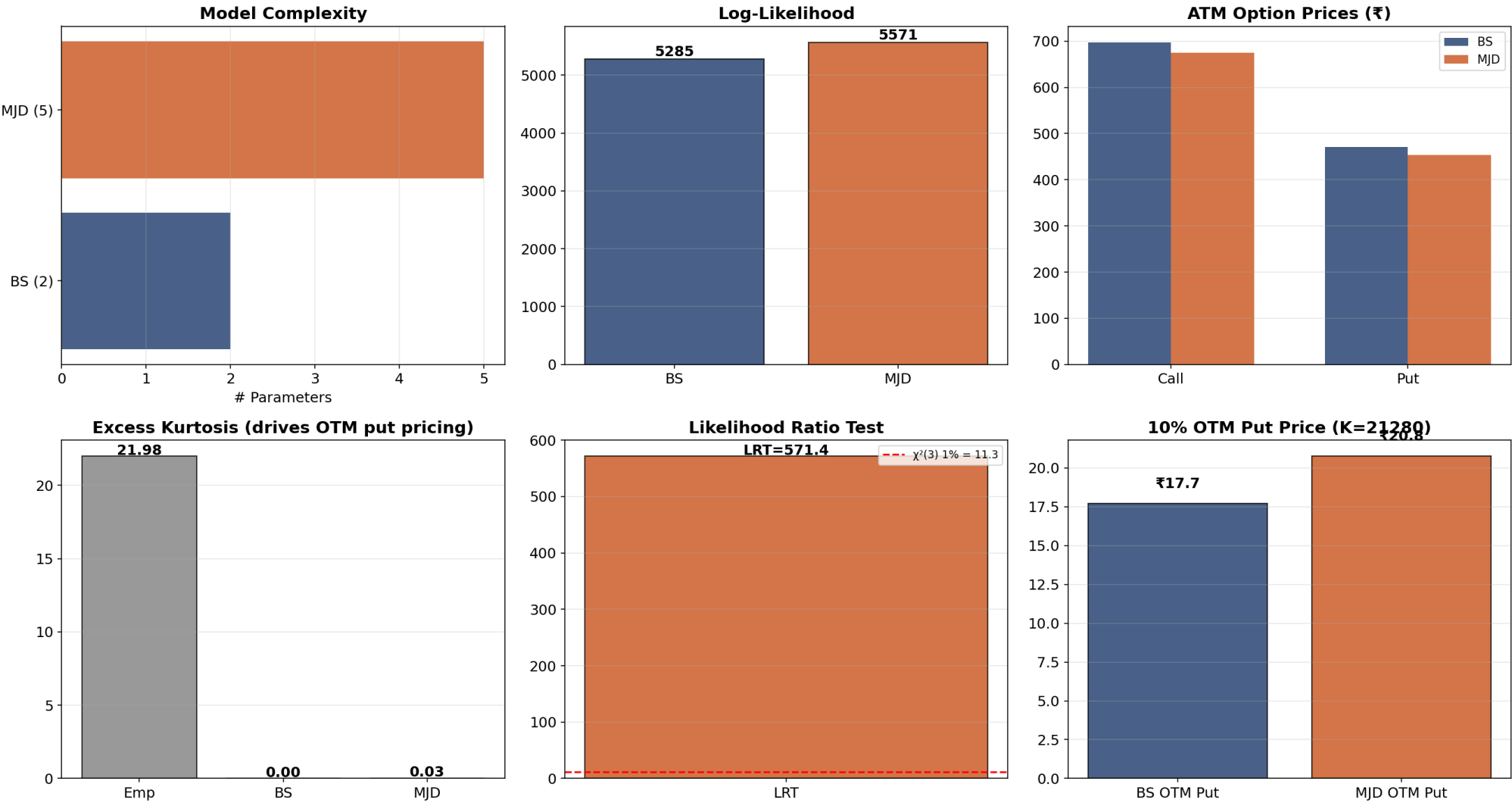


QQ: Empirical vs MJD — better tail fit



Summary: BS vs MJD Option Pricing Comparison

BS vs MJD: Comprehensive Option Pricing Comparison on NIFTY 50



Model Comparison for Option Pricing

Feature	Black-Scholes	Merton JD
Underlying SDE	GBM (continuous)	GBM + Poisson jumps
Parameters	2 (μ , σ)	5 (μ , σ , λ , μ_J , σ_J)
Return distribution	Gaussian	Gaussian mixture
Option pricing	Closed-form (BS formula)	Monte Carlo required
Implied volatility	Flat (constant σ)	Smile / skew
OTM put pricing	Systematically low	More realistic
Captures crashes	No	Yes (jump component)
Calibration	Analytic MLE	Numerical MLE (L-BFGS-B)

7. Future Work & Connection to Supervisor Research

End-term: extend option pricing comparison to stochastic volatility models:

- Heston (1993): variance $v(t)$ follows CIR process $\rightarrow$ produces full IV surface
- Bates (1996): Heston + jumps $\rightarrow$ 8 parameters, most flexible model
- Calibrate to live NIFTY option chain prices (not just returns)

Numerical challenge: CIR process has non-Lipschitz $\sqrt{v(t)}$ coefficient

$\rightarrow$ Standard Euler-Maruyama and Milstein Schemes will blow up, need to use tamed Milstein Schemes.

Prof. Chaman Kumar's research directly addresses this:

- Tamed Euler for Lévy SDEs [DKS16, SIAM J. Numer. Anal., 54(3)]
- Tamed Milstein for super-linear coefficients [KS17, EJP 22]
- Milstein for Markovian switching [KK20, J. Comp. Appl. Math. 377]
- McKean-Vlasov equations [KNRS20] — future scope: mean-field option markets

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Thank You

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Questions?